

Name Class Date **Solano's 4th grade Number Patterns**

Total questions: 10

Worksheet time: 13mins

1. What reference are we using for the input and output table?

a) A soda machine

b) A television

c) Shoes

d) A transformer

2. What never changes?

a) The rule

b) The output

c) the input

d) the pattern

3.

|  |  |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| 10                                                                                 | 15                                                                                 |
| 15                                                                                 | 20                                                                                 |
| 20                                                                                 | 25                                                                                 |
| 25                                                                                 | ?                                                                                  |

What is the output?

a) 35

b) 40

c) 30

d) 50

4.

|  |  |
|----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 900                                                                              | 800                                                                               |
| 700                                                                              | 600                                                                               |
| 500                                                                              | ?                                                                                 |
| 300                                                                              | 200                                                                               |

What is the output?

a) 300

b) 600

c) 100

d) 400

5.

|  |  |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
| 10                                                                                  | 21                                                                                  |
| 12                                                                                  | 23                                                                                  |
| 13                                                                                  | ?                                                                                   |
| 14                                                                                  | ?                                                                                   |

What are the correct values that goes in the output?

a) 22

b) 24

c) 10

d) 25

6.

|  |  |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| 550                                                                               | 50                                                                                |
| 450                                                                               | 400                                                                               |
| 350                                                                               | ?                                                                                 |
| 150                                                                               | ?                                                                                 |

What are the correct values that goes in the output?

- [illegible]

7.

|  | Rule |  |
|-------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------|
| 17                                                                                  | ?    | 24                                                                                  |
| 18                                                                                  | ?    | 25                                                                                  |
| 19                                                                                  | ?    | 26                                                                                  |
| 27                                                                                  | ?    | 34                                                                                  |
| 45                                                                                  | ?    | 52                                                                                  |

What it is the rule (Number expression) in this input and output table?

- [illegible]

8.

|                                                                                  |                                                                                  |
|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|  |  |
| 1                                                                                | 9                                                                                |
| 3                                                                                | 11                                                                               |
| 5                                                                                | 13                                                                               |
|                                                                                  | 15                                                                               |
| 9                                                                                |                                                                                  |

Fill in the input and output table correctly.

a) Input 6

b) Input 7

c) Output 17

d) Output 16

9.

34 The table shows a relationship between input numbers and output numbers.

Number Machine

| Input | Output |
|-------|--------|
| 1     | 10     |
| 2     | 11     |
| 3     | 12     |
| 4     | 13     |

Which rule can be used to find the output number when the input number is given?

Select the best answer.

a) -9

b)  $\times 10$ c)  $\times 4$ 

d) +9

10.

The table shows the relationship between the position of a number in a pattern and its value.

| Position | Value |
|----------|-------|
| 1        | 33    |
| 2        | 34    |
| 3        | 35    |
| 4        | 36    |

Which rule shows how to find the value when given the position?

Select the best answer.

a)  $\times 33$

b)  $-32$

c)  $\div 33$

d)  $+32$